Pepper Diseases

Identification, causes, favourable conditions, management and verified remedies

Seven diseases of pepper (Capsicum annuum: capsicum / bell pepper and chilli), India focus, including bacterial spot

Primary sources: TNAU Agritech Portal (chilli), ICAR-IARI and Himachal Pradesh studies on bacterial spot, CIB&RC (Insecticides Act, 1968) Major Uses of Pesticides up to 30.09.2025, and supplementary international sources. Compiled 21 October 2026.

Contents

About this document

Summary table

1. Bacterial Spot (Bacterial Leaf Spot)
2. Anthracnose, Fruit Rot and Die-back
3. Powdery Mildew
4. Cercospora Leaf Spot (Frog-eye)
5. Damping-off (Nursery)
6. Fusarium Wilt (Root Wilt)
7. Chilli Leaf Curl and Other Viral Diseases

Appendix A. Cross-check log and gaps

Appendix B. Image-classification guide for the model classes

About this document

Scope and method

Each disease chapter follows the same six headings: identification; causes and progression; favourable conditions; prevention and management; remedies and treatment; and sources. Chemical recommendations are included only where they could be matched to the CIB&RC list of registered chilli uses or to a published TNAU, ICAR or university recommendation, and each is tagged to show how it was verified.

Verification tags used in the chemical tables

V1 - Verified against CIB&RC. The product is listed for CHILLI (Capsicum annuum) and this disease (or a named group including it) in the CIB&RC 'Major Uses of Pesticides' fungicide or bio-pesticide compilation, up to 30.09.2025. Doses are per hectare as printed.

V2 - TNAU, ICAR or university publication only, or a generic CIB&RC claim ('leaf spot') that does not name this disease. Confirm current registration and the product label.

-- - No chemical option verified for this disease; none is recommended.

Important cautions

- The CIB&RC compilation is 'based on certificate issued' up to 30.09.2025 and carries a disclaimer that it is for guidance and not for legal purposes. A hosted copy was read, not the live register. Check the

current label and the CIB&RC register (ppqs.gov.in) and follow the pre-harvest interval (PHI) on the label.

- The CIB&RC rows name the crop 'chilli'. Capsicum or bell pepper is not named in the rows read, so confirm the label before using any product on capsicum, especially under protected cultivation. 'Pepper' here means Capsicum annuum (capsicum / bell pepper and chilli); black pepper (Piper nigrum) is a different crop and is not covered.
- Doses are per hectare as printed in CIB&RC; TNAU and trial advice are given as concentrations. Chilli and capsicum are harvested repeatedly, so pick products and sprays that fit the PHI. Rotate chemical groups, use PPE and follow label safety instructions.
- Disease descriptions rest mainly on TNAU Agritech pages for chilli and ICAR-linked Indian studies of bacterial leaf spot. Sources marked supplementary (tomato pages, Australian, Texan, Brazilian, Plantix and educational sources) support only the points where they are cited. No MPKV Rahuri, PDKV Akola or UC IPM pepper page was retrieved.

Summary table

Disease	Pathogen / type	Main non-chemical measures	Chemical status
Bacterial Spot	Xanthomonas euvesicatoria (bacterial; seed- and debris-borne)	Clean seed, resistant lines, early preventive sprays, sanitation	V1: kasugamycin 3% SL; streptocycline 9+1 SP; copper generic (V2)
Anthracnose, Fruit Rot and Die-back	Colletotrichum capsici (seed-borne; rain-dispersed)	Clean seed, alternate chemical groups	Many V1 options (CIB&RC chilli)
Powdery Mildew	Leveillula taurica (obligate parasite)	Scout from flowering; spray at first appearance	Many V1 options; sulphur and others
Cercospora Leaf Spot	Cercospora capsici (residue-borne)	Spacing, drip irrigation, early scouting	V1: kasugamycin, tebuconazole 38.39% SC, a three-way mixture
Damping-off	Pythium aphanidermatum (soil- and seed-borne)	Raised beds, drainage, seed treatment	V1 seed dressings and bio-agents
Fusarium Wilt	Fusarium oxysporum f. sp. capsici (soil-borne)	No cultural measure sourced	V1 Trichoderma viride only
Leaf Curl and Viral Diseases	Chilli leaf curl virus (whitefly-borne)	Nursery nets, rogueing, tolerant varieties	None; fungicides do not apply

Quick separation of look-alikes: small water-soaked leaf spots with a yellow halo, leaf drop and scabby fruit spots point to bacterial spot; brown circular leaf spots with light grey centres and dark margins point to Cercospora; small black sunken spots on ripe fruit with bleached pods and twig die-back point to anthracnose; yellow upper-leaf patches with white powder beneath point to powdery mildew; seedlings collapsing in the nursery point to damping-off; yellowing, wilting plants with discoloured vascular tissue point to Fusarium wilt; and upward-curled, thickened, stunted plants point to leaf curl virus.

1. Bacterial Spot (Bacterial Leaf Spot)

Pepper (Capsicum annuum: capsicum / bell pepper and chilli) | Xanthomonas euvesicatoria (older name X. campestris pv. vesicatoria) | Bacterial (seed- and debris-borne; rain-splash)

At a glance. The main bacterial disease of capsicum and chilli: small water-soaked leaf spots with yellow halos, leaf drop and raised scabby fruit spots in warm, humid, rainy weather. Clean seed is the first defence and protection works best when started early. For chilli, CIB&RC lists kasugamycin 3% SL and streptocycline (9% + 1% SP) for bacterial leaf spot. Copper alone gives limited control.

1. Identification

Symptoms

- Crop / stage: leaves, stems and fruit of capsicum and chilli, in the nursery and the field (TXP, IARI).
- Leaves: small, yellowish-green to brown spots that become numerous and sometimes coalesce into large spots; infected leaves turn yellow and fall (TXP). TNAU's tomato page describes the same pathogen group as small, brown, water-soaked, circular spots with a yellow halo and defoliation of older leaves (TNT).
- Spots can turn blackish and the dead tissue may drop out, leaving a hole in the leaf (TXT, tomato).
- Fruit: small water-soaked spots on green fruit whose centres become irregular, light brown, sunken and scabby (TNT); black raised specks that later become scab-like spots are described on tomato fruit (TXT).
- The disease was found in moderate to severe form in bell pepper areas of Solan, Shimla and Sirmour districts of Himachal Pradesh (GP).

Distinguishing features

- Cercospora leaf spot lesions are brown and circular with light grey centres and dark brown margins (TP5), unlike small water-soaked, yellow-haloed bacterial spots.
- Anthracnose shows small black circular sunken spots on ripe fruit, with straw-coloured, bleached fruit when severe (TPH).
- Leaf curl virus causes curling, vein swelling and stunting rather than spots (UPL).

2. Causes and progression

- Pathogen: in a study of 31 isolates from tomato, chilli and bell pepper in Himachal Pradesh, Karnataka, Uttarakhand, Tamil Nadu and Haryana, all were identified as Xanthomonas euvesicatoria (IARI). A bell pepper study in Himachal Pradesh named the pathogen X. campestris pv. vesicatoria (GP), the older name.
- Four Xanthomonas species are associated with the disease worldwide; X. euvesicatoria essentially infects capsicum, chilli and tomato (AUS).
- Survival and spread: the disease is seed- and soil-borne, infected seed is the source of infection for emerging seedlings, and splashing rain spreads the bacterium from diseased to healthy plants (TXP). Infected debris and seed are the primary source and rain splash the secondary spread (TNT). Seed transmissibility is high (AUS).
- The pathogen's optimum growth was at 30 C and pH 7.0 in culture (GP).

3. Favourable conditions

- Moderate temperatures (about 25 C) with high relative humidity (about 80%) favoured disease development in Himachal Pradesh studies (GP).
- Moist weather and splattering rain, high humidity or persistent dew (TNT).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use disease-free seed and healthy transplants; seed is the main route into new fields (TNT, TXP, AUS). An Australian project recommends heat treatment of seed to prevent primary introduction and reduce the risk of new races (conditions are in the project report) (AUS).
- Use resistant capsicum lines where possible; most common green pepper types are susceptible (AUS, TXP).
- Start a preventive spray programme early in the season and continue it at regular intervals (TXP).
- Rotate crops and clear plant debris promptly; rotation and careful transplant selection are advised for the same pathogen on tomato (TXT, supplementary).

Regional note

No Maharashtra-specific recommendation was found. Indian studies cover Himachal Pradesh, Karnataka, Uttarakhand, Tamil Nadu and Haryana (IARI, GP).

5. Remedies and treatment

Biological and non-registered options

- Pseudomonas fluorescens was the only one of five biocontrol agents that inhibited the bacterium in vitro in the Himachal Pradesh study (GP). The CIB&RC bio-fungicide list has no pepper bacterial spot entry (CIBB); P. fluorescens is listed there for chilli damping-off, not bacterial spot.

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Kasugamycin 3% SL	Bacterial leaf spot, Cercospora leaf spot and die-back (chilli)	1000-1250 ml/ha (30-37.5 g a.i.)	375-500 L	V1	CIB&RC (chilli); PHI 5 d
Streptomycin sulphate 9% + Tetracycline HCl 1% SP (Streptocycline)	Bacterial leaf spot (label: tomato / chillies)	Spray 40-100 ppm on seedlings in seed beds and the field after the first true leaves appear; two sprays, one before transplanting and one after (TNAU tomato advice: 40-100 ppm, 20 g/acre)	-	V1	CIB&RC (tomato/chillies entry for the 9% + 1% product; the 90% + 10% entry names tomato only); PHI not printed; TNT
Copper oxychloride 50% WP	Chilli 'leaf spot' (bacterial spot not named)	2.5 kg/ha	750-1000 L	V2	CIB&RC generic leaf spot claim; PHI not stated; copper alone is weak against copper-tolerant Xanthomonas (AUS)
Tribasic copper sulfate 34.5% SC	Chilli 'leaf spot' and anthracnose (bacterial spot not named)	2000 ml/ha	500 L	V2	CIB&RC generic leaf spot claim; PHI 5 d
Streptomycin sulphate 0.01% + Copper oxychloride 0.25% (foliar)	Bacterial spot of bell pepper	Foliar sprays; streptomycin 0.01% alone was also effective in the trial	-	V2	Himachal Pradesh thesis (GP); research result, minimum disease and highest yield

CIB&RC's chilli entries are for 'chilli' (Capsicum annuum). The capsicum or bell pepper use is not named in the rows read, so confirm the label before using these products on capsicum, particularly under protected cultivation.

Copper tolerance was found in all Australian isolates tested, and copper alone controlled the disease poorly (AUS); a Brazilian tomato study found streptomycin-resistant strains in some Xanthomonas groups (BRZ, supplementary). Rotate with non-copper products and do not rely on one chemistry.

Dose units follow the CIB&RC list; the label governs. PHI for streptocycline is not printed in the entry read.

6. Sources and citations

[IARI] Characterization of Xanthomonas species causing bacterial leaf spot disease of pepper (Capsicum annuum) in India. Indian Journal of Agricultural Sciences, December 2017 (ICAR-IARI, New Delhi). https://epubs.icar.org.in/index.php/IJAgS/article/download/76514/31945/194030

[GP] Gupta, S.K. & Prashad, D. (2009). Studies on bacterial spot of bell pepper (Capsicum annuum L.) caused by Xanthomonas campestris pv. vesicatoria (Doidge) Dye. Thesis, Himachal Pradesh agricultural university (KRISHI/Krishikosh record). https://krishikosh.egranth.ac.in/signposting/describedby/6a0a9a71-e39d-4392-b26b-f4361f5cb464

[AUS] Hort Innovation (Australia) project VG14010: Management and detection of bacterial leaf spot in capsicum and chilli crops, final report summary (supplementary, non-Indian). https://www.horticulture.com.au/Information-Hub/Our-Projects/Project-Reports/VG14010

[TNT] TNAU Agritech Portal - Tomato: Bacterial leaf spot (Xanthomonas axonopodis pv. vesicatoria); same pathogen group as pepper bacterial spot (supplementary, tomato page). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_11.html

[TXP] Texas A&M Plant Disease Handbook - Pepper (supplementary, non-Indian). https://plantdiseasehandbook.tamu.edu/food-crops/vegetable-crops/pepper

[TXT] Texas A&M AgriLife: Tomato problem solver - bacterial spot (Xanthomonas campestris pv. vesicatoria) (supplementary, tomato). https://aggie-horticulture.tamu.edu/vegetable/problem-solvers/tomato-problem-solver/leaves/bacterial-spot/

[TP5] TNAU Agritech Portal - Chilli: Cercospora leaf spot (Cercospora capsici). https://agritech.tnau.ac.in/crop_protection/chilli_diseases_5.html

[TPH] TNAU Agritech Portal - Post-harvest diseases, chilli: Anthracnose (Colletotrichum capsici). https://agritech.tnau.ac.in/crop_protection/chilli_phdiseases_2.html

[UPL] Leaf curl of chillies (PPA-321 e-content, supplementary educational source). https://www.upcollege.ac.in/Upload/econtent/2475.pdf

[BRZ] Sensitivity to copper, streptomycin and oxytetracycline of xanthomonads associated with bacterial spot in processing tomatoes (Brazil; Horticultura Brasileira; supplementary, tomato). https://www.scielo.br/j/hb/a/ZsKSpzBKsVhpSjBWs4D4Lyr/abstract/?lang=en

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

2. Anthracnose, Fruit Rot and Die-back

Pepper (chilli and capsicum) | Colletotrichum capsici (and related Colletotrichum species) | Fungal (seed-borne; rain- and insect-dispersed conidia)

At a glance. Small black circular sunken spots on ripening fruit, straw-coloured bleached pods and die-back of twigs from the tip. A major constraint to chilli production in India in warm, humid weather with rain at ripening. Clean seed and a rotated spray programme are the core, and CIB&RC lists many products for chilli fruit rot and anthracnose.

1. Identification

Symptoms

- Ripe fruits turning red are affected: small, black, circular spots on the skin; badly affected fruits turn straw-coloured or pale white and lose pungency (TPH).
- Cut fruits show minute elevated sclerotia on the underside of the skin and, later, seeds covered by a mat of hyphae that turns rusty (TPH).
- Leaves show small brown or black water-soaked spots; brown stem spots lead to twig necrosis and die-back that begins at the branch tip (UPA, supplementary).

Distinguishing features

- Bacterial spot gives small water-soaked, yellow-haloed leaf spots and scabby fruit spots rather than sunken black fruit spots with sclerotia (TNT, TPH).
- Cercospora gives frog-eye leaf spots (TP5).

2. Causes and progression

- Pathogen: Colletotrichum capsici; mycelium is septate, acervuli and stroma are hemispherical and conidia appear pinkish in mass (TPH).
- Spread: seed-borne; secondary spread by airborne conidia and rain; flies and other insects carry spores from fruit to fruit (TPH).
- Impact: in a Rajasthan survey, incidence ranged from about 13 to 22% across tehsils of Jodhpur district (KK). Anthracnose and powdery mildew significantly reduced chilli quality and yield in Indian field trials in Karnataka (HAV).

3. Favourable conditions

- Temperature about 28 C and relative humidity about 95%, with high humidity and rain after fruits start to ripen (TPH).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use disease-free seed (TPH).
- TNAU lists seed treatment with thiram or zineb; the doses are printed in units that do not read as seed doses (2 kg/ha and 2.5 kg/ha), so they are not reproduced; follow the label (TPH).
- Alternate chemical groups; products with different modes of action are important for resistance management (HAV).

Regional note

No Maharashtra-specific recommendation was found. Indian data come from Tamil Nadu (TNAU), Rajasthan and Karnataka (KK, HAV).

5. Remedies and treatment

Biological and non-registered options

- TNAU lists biocontrol options for this disease but the text could not be read in full; none is reproduced (TPH). No pepper anthracnose claim exists in the CIB&RC bio-fungicide list (CIBB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Azoxystrobin 23% SC	Fruit rot, powdery mildew	500 ml/ha (125 g a.i.)	500-750 L	V1	CIB&RC (chilli); PHI 5 d
Captan 50% WG	Fruit rot (anthracnose)	1500 g/ha	500 L	V1	CIB&RC (chilli); PHI 5 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Captan 75% WP	Fruit rot	2000 g/ha	1000 L	V1	CIB&RC (chilli); PHI 8 d
Chlorothalonil 75% WP	Fruit rot	800 g/ha	750 L	V1	CIB&RC (chilli); PHI 10 d
Copper oxychloride 50% WP	Leaf spot; fruit rot	2.5 kg/ha	750-1000 L	V1	CIB&RC (chilli); PHI not stated
Copper hydroxide 53.8% DF	Anthracnose	1000 g/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Copper hydroxide 61.41% WG	Anthracnose	875 g/ha	500 L	V1	CIB&RC (chilli); PHI 3 d
Copper sulphate 2.62% SC	Fruit rot (anthracnose)	1 L/ha	500 L	V1	CIB&RC (chilli); PHI 3 d
Difenoconazole 25% EC	Die-back, fruit rot	0.05% (50 ml/100 L)	500-1000 L	V1	CIB&RC (chilli); PHI 15 d
Hexaconazole 2% SC	Powdery mildew and fruit rot	3.0 L/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Hexaconazole 75% WG	Leaf spot, anthracnose, powdery mildew	66.7 g/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Kitazin (IBP) 48% EC	Fruit rot / die-back	0.20% (200 ml/100 L)	As required L	V1	CIB&RC (chilli); PHI 3 d
Kresoxim-methyl 44.3% SC	Powdery mildew, fruit rot, die-back, twig blight	500 ml/ha (250 g a.i.)	500 L	V1	CIB&RC (chilli); PHI 25 d
Mancozeb 75% WP	Fruit rot, ripe rot, leaf spot	1.5-2 kg/ha	750 L	V1	CIB&RC (chilli); PHI not stated
Metiram 70% WG	Anthracnose	2000 g/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Pyraclostrobin 20% WG	Anthracnose	500 g/ha	500 L	V1	CIB&RC (chilli); PHI 3 d
Tebuconazole 25.9% EC	Fruit rot, powdery mildew	0.5-0.75 L/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Tebuconazole 25% WG	Powdery mildew, fruit rot	0.5-0.75 kg/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Thiophanate-methyl 70% WG	Powdery mildew, die-back, anthracnose	715 g/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Thiophanate-methyl 41.7% SC	Fruit rot / anthracnose	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Tribasic copper sulfate 34.5% SC	Leaf spot and anthracnose	2000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Zineb 75% WP	Fruit rot; leaf spot	1.5-2 kg/ha	750-1000 L	V1	CIB&RC (chilli); PHI not stated
Kasugamycin 3% SL	Bacterial leaf spot, Cercospora leaf spot and die-back	1000-1250 ml/ha (30-37.5 g a.i.)	375-500 L	V1	CIB&RC (chilli); PHI 5 d
Azoxystrobin 25% + Boscalid 35% WG	Powdery mildew and anthracnose	500 g/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Azoxystrobin 18.2% + Difenoconazole 11.4% SC	Anthracnose and powdery mildew	1 ml/L (0.1%)	500 L	V1	CIB&RC (chilli); PHI 5 d
Azoxystrobin 8.3% + Mancozeb 66.7% WG	Powdery mildew, leaf spot, anthracnose	1500 g/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Azoxystrobin 4.7% + Mancozeb 59.7% + Tebuconazole 5.6% WG	Anthracnose, powdery mildew, Choanephora blight, Cercospora leaf spot	2000 g/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Azoxystrobin 30.5% + Myclobutanil 4.85% SE	Powdery mildew and anthracnose	375 (as printed)/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Azoxystrobin 11% + Tebuconazole 18.3% SC	Fruit rot, powdery mildew, die-back	600 ml/ha	500-750 L	V1	CIB&RC (chilli); PHI 5 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Azoxystrobin 12.5% + Tebuconazole 12.5% SC	Powdery mildew and fruit rot	800 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Azoxystrobin 120 g/L + Tebuconazole 240 g/L SC	Fruit rot, die-back, powdery mildew	830 ml/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Azoxystrobin 5.1% + Tebuconazole 9.1% + Prochloraz 18.2% EC	Anthracnose leaf spot, die-back, fruit rot, powdery mildew	1250 ml/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Azoxystrobin 10% + Fipronil 5% SC (fungicide-insecticide)	Powdery mildew; anthracnose / fruit rot	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Captan 70% + Hexaconazole 5% WP	Fruit rot (anthracnose)	500-1000 g/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Carbendazim 12% + Mancozeb 63% WP	Leaf spot, fruit rot, powdery mildew	750 g/ha	500 L	V1	CIB&RC (chilli); PHI 3 d
Carbendazim 25% + Flusilazole 12.5% SE	Powdery mildew, fruit rot, die-back	1000 ml/ha	375-500 L	V1	CIB&RC (chilli); PHI 5 d
Chlorothalonil 40% + Difenoconazole 4% SC	Powdery mildew, anthracnose, fruit rot	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Difenoconazole 10% + Mancozeb 50% WDG	Powdery mildew, die-back, fruit rot	625 g/ha	500 L	V1	CIB&RC (chilli); PHI 13 d
Flubendiamide 3.5% + Hexaconazole 5% WG (insecticide-fungicide)	Leaf spot, powdery mildew, anthracnose	1500 g/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Fluopyram 17.7% + Tebuconazole 17.7% SC	Powdery mildew and anthracnose	500 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Fluopyram 21.37% + Trifloxystrobin 21.37% SC	Powdery mildew and anthracnose	500-600 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Fluxapyroxad 75 g/L + Difenoconazole 50 g/L SC	Anthracnose and powdery mildew	800-1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Fluxapyroxad 250 g/L + Pyraclostrobin 250 g/L SC	Powdery mildew, anthracnose	200-250 ml/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Kresoxim-methyl 15% + Chlorothalonil 56% WG	Powdery mildew, leaf spot, anthracnose	1000 g/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Mefentrifluconazole 200 g/L + Pyraclostrobin 200 g/L SC	Anthracnose and Choanephora blight	400-500 ml/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Mancozeb 50% + Thiophanate-methyl 25% WG	Leaf spot, fruit rot, powdery mildew	1500 g/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Picoxystrobin 6.78% + Tricyclazole 20.33% SC	Anthracnose, wet rot, powdery mildew	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 3 d
Prochloraz 24.4% + Tebuconazole 12.1% EW	Fruit rot, die-back, powdery mildew	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Pydiflumetofen 6.89% + Difenoconazole 11.49% SC	Powdery mildew, leaf spot, fruit rot, die-back	500 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Pyraclostrobin 10% + Fipronil 5% SC (fungicide-insecticide)	Powdery mildew; anthracnose / fruit rot	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Sulphur 58.5% + Azoxystrobin 4.2% SC	Powdery mildew, anthracnose	2000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Tebuconazole 10% + Sulphur 65% WG	Powdery mildew and fruit rot	1250 g/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Tebuconazole 50% + Trifloxystrobin 25% WG	Powdery mildew, anthracnose, Alternaria leaf spot	250 g/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Tebuconazole 15% + Zineb 57% WDG	Powdery mildew, leaf spot, Choanephora blight, anthracnose fruit rot	1250 g/ha	500 L	V1	CIB&RC (chilli); PHI 28 d
Tebuconazole 6.7% + Captan 26.9% SC	Powdery mildew and anthracnose	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Trifloxystrobin 25% + Hexaconazole 25% WG	Leaf spot, powdery mildew, anthracnose	200 g/ha	500 L	V1	CIB&RC (chilli); PHI 7 d

The label crop name is 'chilli'; confirm the label for capsicum. Several products carry insecticide partners (fipronil, flubendiamide); use these only if the insect target is also present.

In a Rajasthan trial, a tebuconazole + trifloxystrobin spray gave the lowest anthracnose severity (11.55%) (KK); this combination is labelled for chilli anthracnose (250 g/ha, PHI 5 d). Rotate chemical groups and follow each PHI, because chilli is picked repeatedly.

6. Sources and citations

[TPH] TNAU Agritech Portal - Post-harvest diseases, chilli: Anthracnose (Colletotrichum capsici). https://agritech.tnau.ac.in/crop_protection/chilli_phdiseases_2.html

[UPA] Anthracnose and fruit rot of chilli (PPA-321 e-content, supplementary educational source). https://www.upcollege.ac.in/Upload/econtent/2474.pdf

[KK] Studies on anthracnose (Colletotrichum capsici) of chilli and its management (Rajasthan; Krishikosh thesis record). https://krishikosh.egranth.ac.in/items/2cd28225-4b00-422d-923f-9fc19637e613

[HAV] Novel management strategy to minimize the growing threat of fruit rot and powdery mildew diseases of chilli in India. Journal of Pharmacognosy and Phytochemistry 9(6): 2250-2255 (2020; Karnataka). https://phytojournal.com/archives/2020.v9.i6.13626/novel-management-strategy-to-minimize-the-growing-threat-of-fruit-rot-and-powdery-mildew-diseases-of-chilli-capsicum-annuum-in-india

[TNT] TNAU Agritech Portal - Tomato: Bacterial leaf spot (Xanthomonas axonopodis pv. vesicatoria); same pathogen group as pepper bacterial spot (supplementary, tomato page). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_11.html

[TP5] TNAU Agritech Portal - Chilli: Cercospora leaf spot (Cercospora capsici). https://agritech.tnau.ac.in/crop_protection/chilli_diseases_5.html

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

3. Powdery Mildew

Pepper (chilli and capsicum) | Leveillula taurica (anamorph Oidiopsis taurica) | Fungal (obligate parasite; internal mycelium, spores through stomata)

At a glance. Yellow patches on the upper leaf surface and white powdery growth underneath, leading to leaf shedding. A serious and growing problem for chilli in India. Sprays at first appearance are the main control; CIB&RC lists the largest number of products of any chilli disease for this one.

1. Identification

Symptoms

- Leaves: shedding of foliage and white powdery growth on the lower side of leaves (TP3). On pepper, yellow areas that may turn brown appear on the upper surface above the white powder on the underside (TXP).
- Repeated spore cycles can cause severe outbreaks that damage the crop economically (AGR, supplementary).

Distinguishing features

- Powdery mildew is the only disease in this document with white powdery growth on the leaf underside (TP3, TXP).
- Leaf curl virus causes curling and vein swelling, not powdery growth (UPL).

2. Causes and progression

- Pathogen: Leveillula taurica, with Oidiopsis taurica as the anamorph (AGR). Spores germinate on the leaf and the fungus grows into the leaf; spores appear on the underside, with a latency of 18 to 21 days in one summary (AGR, supplementary).
- Impact: powdery mildew and anthracnose together are described as significantly reducing chilli quality and yield in India, especially in recent years (HAV).

3. Favourable conditions

- Weather thresholds were not given in the TNAU page or the other Indian sources reviewed.

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Scout from flowering and start sprays at first appearance (TP3).
- Rotate fungicide groups; the Karnataka trial stresses products with novel modes of action for resistance management (HAV).

Regional note

No Maharashtra-specific recommendation was found.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Azoxystrobin 23% SC	Fruit rot, powdery mildew	500 ml/ha (125 g a.i.)	500-750 L	V1	CIB&RC (chilli); PHI 5 d
Bupirimate 26.7% w/w EC	Powdery mildew	2000 ml/ha (500 g a.i.)	500 L	V1	CIB&RC (chilli); PHI 7 d
Cyflufenamid 5% EW	Powdery mildew (Leveillula taurica)	300 ml/ha	375-500 L	V1	CIB&RC (chilli); PHI 5 d
Hexaconazole 2% SC	Powdery mildew and fruit rot	3.0 L/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Hexaconazole 75% WG	Leaf spot, anthracnose, powdery mildew	66.7 g/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Kresoxim-methyl 44.3% SC	Powdery mildew, fruit rot, die-back, twig blight	500 ml/ha (250 g a.i.)	500 L	V1	CIB&RC (chilli); PHI 25 d
Meptyl dinocap 35.7% EC	Powdery mildew	308.6-342.8 ml/ha	500-750 L	V1	CIB&RC (chilli); PHI 7 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Metrafenone 500 g/L SC	Powdery mildew	200-250 ml/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Polyoxin D zinc salt 5% SC	Powdery mildew	1000 (as printed; g or ml)/ha	500 L	V1	CIB&RC (chilli); PHI 0 d
Sulphur 52% FS	Powdery mildew	2.00 L/ha (1.04 kg a.i.)	400 L	V1	CIB&RC (chilli); PHI not stated
Sulphur 80% WP	Powdery mildew (chillies and okra)	3.13 kg/ha	750-1000 L	V1	CIB&RC (chilli); PHI not stated
Sulphur 80% WG	Powdery mildew	3.13 kg/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Tebuconazole 25.9% EC	Fruit rot, powdery mildew	0.5-0.75 L/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Tebuconazole 25% WG	Powdery mildew, fruit rot	0.5-0.75 kg/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Tebuconazole 38.39% SC	Powdery mildew and Cercospora leaf spot	440-500 ml/ha	375-500 L	V1	CIB&RC (chilli); PHI 5 d
Thiophanate-methyl 70% WG	Powdery mildew, die-back, anthracnose	715 g/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Azoxystrobin 25% + Boscalid 35% WG	Powdery mildew and anthracnose	500 g/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Azoxystrobin 18.2% + Difenoconazole 11.4% SC	Anthracnose and powdery mildew	1 ml/L (0.1%)	500 L	V1	CIB&RC (chilli); PHI 5 d
Azoxystrobin 8.3% + Mancozeb 66.7% WG	Powdery mildew, leaf spot, anthracnose	1500 g/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Azoxystrobin 4.7% + Mancozeb 59.7% + Tebuconazole 5.6% WG	Anthracnose, powdery mildew, Choanephora blight, Cercospora leaf spot	2000 g/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Azoxystrobin 30.5% + Myclobutanil 4.85% SE	Powdery mildew and anthracnose	375 (as printed)/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Azoxystrobin 11% + Tebuconazole 18.3% SC	Fruit rot, powdery mildew, die-back	600 ml/ha	500-750 L	V1	CIB&RC (chilli); PHI 5 d
Azoxystrobin 12.5% + Tebuconazole 12.5% SC	Powdery mildew and fruit rot	800 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Azoxystrobin 120 g/L + Tebuconazole 240 g/L SC	Fruit rot, die-back, powdery mildew	830 ml/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Azoxystrobin 5.1% + Tebuconazole 9.1% + Prochloraz 18.2% EC	Anthracnose leaf spot, die-back, fruit rot, powdery mildew	1250 ml/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Azoxystrobin 10% + Fipronil 5% SC (fungicide-insecticide)	Powdery mildew; anthracnose / fruit rot	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Boscalid 25.2% + Pyraclostrobin 12.8% WG	Powdery mildew	600 g/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Carbendazim 12% + Mancozeb 63% WP	Leaf spot, fruit rot, powdery mildew	750 g/ha	500 L	V1	CIB&RC (chilli); PHI 3 d
Carbendazim 25% + Flusilazole 12.5% SE	Powdery mildew, fruit rot, die-back	1000 ml/ha	375-500 L	V1	CIB&RC (chilli); PHI 5 d
Chlorothalonil 40% + Difenoconazole 4% SC	Powdery mildew, anthracnose, fruit rot	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Difenoconazole 10% + Mancozeb 50% WDG	Powdery mildew, die-back, fruit rot	625 g/ha	500 L	V1	CIB&RC (chilli); PHI 13 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Flubendiamide 3.5% + Hexaconazole 5% WG (insecticide-fungicide)	Leaf spot, powdery mildew, anthracnose	1500 g/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Fluopyram 17.7% + Tebuconazole 17.7% SC	Powdery mildew and anthracnose	500 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Fluopyram 21.37% + Trifloxystrobin 21.37% SC	Powdery mildew and anthracnose	500-600 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Fluxapyroxad 75 g/L + Difenoconazole 50 g/L SC	Anthracnose and powdery mildew	800-1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Fluxapyroxad 250 g/L + Pyraclostrobin 250 g/L SC	Powdery mildew, anthracnose	200-250 ml/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Kresoxim-methyl 15% + Chlorothalonil 56% WG	Powdery mildew, leaf spot, anthracnose	1000 g/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Mancozeb 50% + Thiophanate-methyl 25% WG	Leaf spot, fruit rot, powdery mildew	1500 g/ha	500 L	V1	CIB&RC (chilli); PHI 10 d
Picoxystrobin 6.78% + Tricyclazole 20.33% SC	Anthracnose, wet rot, powdery mildew	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 3 d
Prochloraz 24.4% + Tebuconazole 12.1% EW	Fruit rot, die-back, powdery mildew	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Pydiflumetofen 6.89% + Difenoconazole 11.49% SC	Powdery mildew, leaf spot, fruit rot, die-back	500 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Pyraclostrobin 10% + Fipronil 5% SC (fungicide-insecticide)	Powdery mildew; anthracnose / fruit rot	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Sulphur 58.5% + Azoxystrobin 4.2% SC	Powdery mildew, anthracnose	2000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Tebuconazole 10% + Sulphur 65% WG	Powdery mildew and fruit rot	1250 g/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Tebuconazole 50% + Trifloxystrobin 25% WG	Powdery mildew, anthracnose, Alternaria leaf spot	250 g/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Tebuconazole 15% + Zineb 57% WDG	Powdery mildew, leaf spot, Choanephora blight, anthracnose fruit rot	1250 g/ha	500 L	V1	CIB&RC (chilli); PHI 28 d
Tebuconazole 6.7% + Captan 26.9% SC	Powdery mildew and anthracnose	1000 ml/ha	500 L	V1	CIB&RC (chilli); PHI 5 d
Trifloxystrobin 25% + Hexaconazole 25% WG	Leaf spot, powdery mildew, anthracnose	200 g/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Wettable sulphur 3 g/L, or carbendazim 1 g/L	Powdery mildew	Three sprays at 15-day intervals from first appearance	-	V2	TNAU (TP3); no carbendazim 50 WP claim for chilli powdery mildew in the rows read
Wettable sulphur 0.25%, or dinocap 0.05%	Powdery mildew	As stated	-	V2	TNAU (TP3); dinocap is not on the CIB&RC chilli rows (meptyl dinocap is)

Label doses are per hectare; TNAU's are concentrations. Sulphur can injure plants in hot weather; follow the label. Keep to each PHI; chilli is harvested repeatedly.

6. Sources and citations

[TP3] TNAU Agritech Portal - Chilli: Powdery mildew (Leveillula taurica). https://agritech.tnau.ac.in/crop_protection/chilli_diseases_3.html

[TXP] Texas A&M Plant Disease Handbook - Pepper (supplementary, non-Indian). https://plantdiseasehandbook.tamu.edu/food-crops/vegetable-crops/pepper

[AGR] Agridots: Diseases of chilli (educational compilation; supplementary, non-primary source). https://agridots.com/courses/bsc-agriculture/plant-pathology/path371-disease-of-horticultural-crops-and-their-management/08-diseases-of-chilli

[HAV] Novel management strategy to minimize the growing threat of fruit rot and powdery mildew diseases of chilli in India. Journal of Pharmacognosy and Phytochemistry 9(6): 2250-2255 (2020; Karnataka). https://phytojournal.com/archives/2020.v9.i6.13626/novel-management-strategy-to-minimize-the-growing-threat-of-fruit-rot-and-powdery-mildew-diseases-of-chilli-capsicum-annuum-in-india

[UPL] Leaf curl of chillies (PPA-321 e-content, supplementary educational source). https://www.upcollege.ac.in/Upload/econtent/2475.pdf

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

4. Cercospora Leaf Spot (Frog-eye)

Pepper (chilli and capsicum) | Cercospora capsici | Fungal (residue-borne; rain- and splash-dispersed)

At a glance. Brown circular leaf spots with light grey centres and dark brown margins that can reach 1 cm or more, causing premature leaf drop and lower yield. Spray at first appearance; CIB&RC names Cercospora for three chilli products and lists others under 'leaf spot'.

1. Identification

Symptoms

- Leaf lesions are brown and circular with small to large light-grey centres and dark brown margins; they may enlarge to 1 cm or more and sometimes coalesce (TP5).
- Stem, petiole and pod lesions have light grey centres with dark borders but are typically elliptical (TP5).
- Severely infected leaves drop prematurely, reducing yield; the white centre often falls out leaving a shot-hole look, and leaf loss exposes fruit to sunscald (TP5, PX1).

Distinguishing features

- Bacterial spot lesions are small, water-soaked and yellow-haloed, without a grey centre (TNT).
- Anthracnose gives sunken black fruit lesions (TPH).

2. Causes and progression

- Pathogen: Cercospora capsici, resilient in the tropics and affecting plants in seedbeds and fields (PX1, supplementary).
- Spread: by water, rain, wind, leaf-to-leaf contact and on tools and workers; foliar infection is by direct penetration, favoured by prolonged leaf wetness (PX1).

3. Favourable conditions

- Prolonged leaf wetness (PX1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Space plants for air circulation, use drip irrigation to reduce leaf wetness and check seedbeds and young plants for early symptoms (PX1).

Regional note

No Maharashtra-specific recommendation was found.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Tebuconazole 38.39% SC	Powdery mildew and Cercospora leaf spot	440-500 ml/ha	375-500 L	V1	CIB&RC (chilli); PHI 5 d
Kasugamycin 3% SL	Bacterial leaf spot, Cercospora leaf spot and die-back	1000-1250 ml/ha (30-37.5 g a.i.)	375-500 L	V1	CIB&RC (chilli); PHI 5 d
Azoxystrobin 4.7% + Mancozeb 59.7% + Tebuconazole 5.6% WG	Anthracnose, powdery mildew, Choanephora blight, Cercospora leaf spot	2000 g/ha	500 L	V1	CIB&RC (chilli); PHI 7 d
Mancozeb 75% WP	Chilli 'leaf spot' (Cercospora not named)	1.5-2 kg/ha (TNAU: 0.25%, two sprays at 10-15 days)	750 L	V2	CIB&RC generic leaf spot claim; TP5
Chlorothalonil (Kavach)	Cercospora leaf spot	0.1%, two sprays at 10-15 days	-	V2	TNAU (TP5); the CIB&RC chilli row for chlorothalonil 75% WP is fruit rot

Apart from kasugamycin, tebuconazole 38.39% SC and the azoxystrobin + mancozeb + tebuconazole mixture, the CIB&RC chilli rows use the generic term 'leaf spot'. Many mixtures list 'leaf spot' for chilli (see the Appendix); confirm the label wording.

6. Sources and citations

[TP5] TNAU Agritech Portal - Chilli: Cercospora leaf spot (Cercospora capsici). https://agritech.tnau.ac.in/crop_protection/chilli_diseases_5.html

[PX1] Plantix crop-disease library: Chilli Cercospora leaf spot (supplementary, non-university source). https://plantix.net/en/library/plant-diseases/100085/chilli-cercospora-leaf-spot/

[TNT] TNAU Agritech Portal - Tomato: Bacterial leaf spot (Xanthomonas axonopodis pv. vesicatoria); same pathogen group as pepper bacterial spot (supplementary, tomato page). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_11.html

[TPH] TNAU Agritech Portal - Post-harvest diseases, chilli: Anthracnose (Colletotrichum capsici). https://agritech.tnau.ac.in/crop_protection/chilli_phdiseases_2.html

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

5. Damping-off (Nursery)

Pepper (chilli and capsicum) | Pythium aphanidermatum | Oomycete (soil- and seed-borne)

At a glance. Seedlings rot before or after emergence and topple, in wet, poorly drained, overcrowded nurseries. Chilli nursery losses of more than 60% have been reported. Seed treatment and raised, well-drained beds are the foundation; CIB&RC lists several seed dressings and bio-agents for chilli.

1. Identification

Symptoms

- Seedlings are killed before emergence, or water-soaked, shrivelled stems collapse after emergence; affected seedlings look light brown and lodge (TP1).
- Germination is reduced and the disease appears soon after sowing in seed beds (TP1).

Distinguishing features

- Fusarium wilt appears in older plants as yellowing and wilting with discoloured vascular tissue (AGR).
- Bacterial spot starts as leaf spots on older seedlings and plants, not stem collapse (TNT).

2. Causes and progression

- Pathogen: Pythium aphanidermatum; hyaline, non-septate mycelium (TP1).
- Spread: through seed and soil (TP1).
- Impact: Pythium damping-off causes more than 60% seedling mortality in nursery and main field in chilli, and is very difficult to manage because of its wide host range and long survival in soil (PYT).

3. Favourable conditions

- Moist soil, poor drainage, relative humidity of 90 to 100% and soil temperature of about 20 C (TP1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Avoid water stagnation, use raised beds, avoid shaded nursery sites, use the recommended seed rate and avoid flooding irrigation (TP1).
- Treat seed with Trichoderma asperellum at 4 g/kg or Pseudomonas fluorescens at 10 g/kg, 24 hours before sowing; apply P. fluorescens to soil at 2.5 kg/ha with 50 kg FYM (TP1).
- In ICAR-IIVR work, thiram, captan, Pseudomonas fluorescens and Trichoderma viride gave good control (IIVR).

Regional note

No Maharashtra-specific recommendation was found. Indian data come from TNAU and ICAR-IIVR (TP1, IIVR).

5. Remedies and treatment

Biological and non-registered options

- Pseudomonas fluorescens 0.5% WP: chilli seedling damping-off (Pythium aphanidermatum), seed treatment 10 g/kg (CIBB).
- Trichoderma viride 1.0% WP: chilli damping-off, seed treatment 4 g/kg (CIBB).
- Trichoderma viride 5.0% SC: chilli nursery damping-off, seed treatment 2 ml/kg (CIBB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Captan 75% WS	Chilli damping-off (seed)	20-30 g formulation per kg seed (15-25 g a.i.)	-	V1	CIB&RC; seed dresser
Captan 75% WP	Damping-off in nursery incl. chillies (soil drench)	0.25% (2500 g in 1000 L)	1000 L	V1	CIB&RC; soil drench in nursery

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Carbendazim 50% WP	Chilli damping-off (seed)	2 g/kg seed (1 g a.i.)	Wet slurry (printed as '1 Lit./tree', a misprint)	V1	CIB&RC; seed treatment
Metalaxyl-M 31.8% ES	Chilli damping-off (seed dresser)	2.0 ml/kg seed (0.64 g a.i.)	-	V1	CIB&RC; seed dresser
Metalaxyl-M 4% + Mancozeb 64% WP	Chilli nursery damping-off (soil drench)	0.30% (3 g/L), 2.0 L/m2	-	V1	CIB&RC; PHI printed as 53 for the nursery drench
Pseudomonas fluorescens 0.5% WP; Trichoderma viride 1.0% WP / 5.0% SC	Chilli damping-off (seed)	10 g/kg; 4 g/kg; 2 ml/kg	-	V1	CIB&RC bio-fungicide list (CIBB)
Copper oxychloride 50% WP drench	Damping-off	2.5 g/L at 4 L/sq m	-	V2	TNAU (TP1); the CIB&RC chilli copper row is for leaf spot and fruit rot
Thiram or captan seed treatment	Damping-off	4 g/kg seed	-	V2	TNAU (TP1); the CIB&RC thiram rows read do not name chilli damping-off

Seed dressings carry no PHI. Do not mix a fungicide seed treatment with a bio-agent unless the label allows it.

6. Sources and citations

[TP1] TNAU Agritech Portal - Chilli: Damping off (Pythium aphanidermatum). https://agritech.tnau.ac.in/crop_protection/chilli_diseases_1.html

[IIVR] Saha, S., Naskar, I., Nayak, D.K. & Sarkar, M.K. (2008). Efficacy of seed dressing agents in the control of damping off disease of chilli caused by Pythium aphanidermatum (ICAR-Indian Institute of Vegetable Research; KRISHI record). https://krishi.icar.gov.in/jspui/handle/123456789/65917

[PYT] Exploitation of Trichoderma species on the growth of Pythium aphanidermatum in chilli (Tamil Nadu; PMC3768735). https://pmc.ncbi.nlm.nih.gov/articles/PMC3768735

[TNT] TNAU Agritech Portal - Tomato: Bacterial leaf spot (Xanthomonas axonopodis pv. vesicatoria); same pathogen group as pepper bacterial spot (supplementary, tomato page). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_11.html

[AGR] Agridots: Diseases of chilli (educational compilation; supplementary, non-primary source). https://agridots.com/courses/bsc-agriculture/plant-pathology/path371-disease-of-horticultural-crops-and-their-management/08-diseases-of-chilli

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

6. Fusarium Wilt (Root Wilt)

Pepper (chilli) | Fusarium oxysporum f. sp. capsici | Fungal (soil-borne; vascular wilt)

At a glance. Yellowing and wilting of the upper foliage, followed by death of scattered or whole plants with discoloured vascular tissue. This chapter rests on educational and CIB&RC sources only; a Trichoderma viride product is registered for chilli root wilt.

1. Identification

Symptoms

- Slight initial yellowing of foliage and wilting of the upper leaves; leaves turn yellow and die; a percentage of plants wilt and die, and scattered wilted plants may also occur; the vascular system is discoloured (AGR, supplementary).

Distinguishing features

- Pythium damping-off kills seedlings in the nursery (TP1); bacterial wilt shows white bacterial streaming from a cut stem in water (not covered in the sources reviewed).

2. Causes and progression

- Pathogen: Fusarium oxysporum f. sp. capsici (AGR).
- Detailed survival and spread information was not found in the sources reviewed.

3. Favourable conditions

- Weather and soil thresholds were not found in the sources reviewed.

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- The only sourced measure is the registered biological option below; no cultural recommendation was found in the sources reviewed.

Regional note

No Maharashtra-specific recommendation was found.

5. Remedies and treatment

Biological and non-registered options

- Trichoderma viride 1.50% WP: chilli root wilt, seed treatment 5 g/kg; seedling dip 5 g/L for 20 minutes; soil application 3.0 kg/ha mixed with 100 kg FYM at transplanting and at flowering (CIBB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Trichoderma viride 1.50% WP	Chilli root wilt	5 g/kg seed; seedling dip 5 g/L for 20 min; 3.0 kg/ha with 100 kg FYM at transplanting and flowering	-	V1	CIB&RC bio-fungicide list (CIBB)
Fungicides or other products	Fusarium wilt (chemical)	-	-	--	No pepper claim for this disease in the CIB&RC list and no verified recommendation; not recommended

No CIB&RC fungicide row for chilli wilt was found; the biological row above is the only registered option read.

6. Sources and citations

[AGR] Agridots: Diseases of chilli (educational compilation; supplementary, non-primary source). https://agridots.com/courses/bsc-agriculture/plant-pathology/path371-disease-of-horticultural-crops-and-their-management/08-diseases-of-chilli

[TP1] TNAU Agritech Portal - Chilli: Damping off (Pythium aphanidermatum). https://agritech.tnau.ac.in/crop_protection/chilli_diseases_1.html

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used:

https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

[CIBB] CIB&RC: 'Major Uses of Bio-Pesticides (Registered under the Insecticides Act, 1968)', updated up to 30.09.2025 (bio-fungicides). Copy used: https://www.jau.in/storage/document/dynamic_document/residue_bio_pesticide_30092025_20251226_780.pdf

7. Chilli Leaf Curl and Other Viral Diseases

Pepper (chilli and capsicum) | Chilli leaf curl virus (a begomovirus, family Geminiviridae) and other viruses | Viral (whitefly-borne; not seed-borne)

At a glance. Upward-curled, thickened, blistered leaves with swollen veins, stunted bushy plants and abscising flower buds, spread by the whitefly Bemisia tabaci. In India it can cause up to 100% yield loss. There is no cure; clean nurseries, whitefly control, rogueing and tolerant varieties are the measures.

1. Identification

Symptoms

- Infected leaves curl upward and roll, veins swell and thicken, interveinal areas show blisters and yellowing, petioles and internodes shorten, young leaves are much reduced and crowded; plants are stunted and bushy, and young flower buds abscise (UPL).
- Symptoms resemble thrips and mite feeding damage (PX2, supplementary).

Distinguishing features

- Thrips or mite damage can look similar (PX2).
- Fungal and bacterial spots do not cause curling of the whole plant (TP5, TNT).

2. Causes and progression

- Pathogen: a begomovirus (Chilli leaf curl virus, Geminiviridae), transmitted by the whitefly Bemisia tabaci in a persistent manner (UPL, PX2).
- The disease is not seed-borne; the virus persists through alternative hosts such as tobacco and tomato and weeds; nursery plants are most prone at the seedling and vegetative stages (PX2, UPL).
- In India, leaf curl is so destructive that it can cause 100% yield loss (UPL, supplementary).

3. Favourable conditions

- Whitefly build-up and infected nursery seedlings (UPL, PX2).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Cover the nursery with nylon nets, monitor regularly, and rogue out diseased plants immediately (PX2, UPL).
- Use resistant or tolerant varieties and seed from virus-free plants (PX2, UPL).
- TNAU's advice for the same type of virus on tomato: raise the nursery under protection, use barrier crops such as sorghum, maize or bajra, yellow sticky traps at 5 per acre, and remove alternate weed hosts (TNL, supplementary).
- Avoid monoculture of chilli and destroy infected plants (AGR).

Regional note

No Maharashtra-specific recommendation was found.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Fungicides (any)	Viral diseases	Not applicable	-	--	Fungicides do not control viruses; no CIB&RC fungicide row applies
Insecticides for whitefly vectors	Vector management	Not verified here	-	--	Registration of whitefly insecticides on chilli or capsicum was not checked in this document; TNAU's tomato page names several, which are not reproduced

Virus control rests on vector management and plant removal; no product cures an infected plant.

6. Sources and citations

[UPL] Leaf curl of chillies (PPA-321 e-content, supplementary educational source). https://www.upcollege.ac.in/Upload/econtent/2475.pdf

[PX2] Plantix crop-disease library: Chilli leaf curl virus (supplementary, non-university source). https://plantix.net/en/library/plant-diseases/200052/chilli-leaf-curl-virus/

[TNL] TNAU Agritech Portal - Tomato: Leaf curl (Tomato leaf curl virus); whitefly-borne begomovirus management (supplementary, tomato page). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_6.html

[AGR] Agridots: Diseases of chilli (educational compilation; supplementary, non-primary source). https://agridots.com/courses/bsc-agriculture/plant-pathology/path371-disease-of-horticultural-crops-and-their-management/08-diseases-of-chilli

[TP5] TNAU Agritech Portal - Chilli: Cercospora leaf spot (Cercospora capsici). https://agritech.tnau.ac.in/crop_protection/chilli_diseases_5.html

[TNT] TNAU Agritech Portal - Tomato: Bacterial leaf spot (Xanthomonas axonopodis pv. vesicatoria); same pathogen group as pepper bacterial spot (supplementary, tomato page). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_11.html

[CIB] Central Insecticides Board & Registration Committee (CIB&RC), Govt. of India: 'Major Uses of Pesticides (Registered under the Insecticides Act, 1968) - Fungicides', up to 30.09.2025, 'based on certificate issued' (Directorate of Plant Protection, Quarantine & Storage). Copy used: https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Earlier edition (31.03.2024): https://ppqs.gov.in/sites/default/files/fungicides_31.03.2024.pdf

Appendix A. Cross-check log and gaps

- Chilli rows in the CIB&RC fungicide list (up to 30.09.2025) were read in full: 66 products and mixtures are tabulated for anthracnose / fruit rot, powdery mildew, Cercospora and leaf spot, plus kasugamycin 3% SL and streptocycline for bacterial leaf spot, and seed or soil products for damping-off.
- Bacterial leaf spot on chilli: kasugamycin 3% SL (PHI 5 d) and streptomycin 9% + tetracycline 1% SP (tomato / chillies entry). The streptocycline 90% + 10% entry names tomato only. No copper product names bacterial spot; copper oxychloride and tribasic copper sulfate carry generic 'leaf spot' claims.
- Products whose chilli claim uses the generic words 'leaf spot': Copper oxychloride 50% WP; Hexaconazole 75% WG; Mancozeb 75% WP; Tribasic copper sulfate 34.5% SC; Zineb 75% WP; Azoxystrobin 8.3% + Mancozeb 66.7% WG; Azoxystrobin 5.1% + Tebuconazole 9.1% + Prochloraz 18.2% EC; Carbendazim 12% + Mancozeb 63% WP; Flubendiamide 3.5% + Hexaconazole 5% WG (insecticide-fungicide); Kresoxim-methyl 15% + Chlorothalonil 56% WG; Mancozeb 50% + Thiophanate-methyl 25% WG; Pydiflumetofen 6.89% + Difenoconazole 11.49% SC; Tebuconazole 50% + Trifloxystrobin 25% WG; Tebuconazole 15% + Zineb 57% WDG; Trifloxystrobin 25% + Hexaconazole 25% WG.
- Items printed in a way that needs care: polyoxin D and some mixtures list doses without units; carbendazim seed treatment lists a dilution as '1 Lit./tree'; metalaxyl-M + mancozeb nursery drench lists a PHI of 53; azoxystrobin + myclobutanil lists '375' without units.
- Bio-fungicide list: Pseudomonas fluorescens and Trichoderma viride products for chilli damping-off and T. viride 1.50% WP for chilli root wilt; no pepper bacterial spot or anthracnose entry.
- No CIB&RC fungicide row exists for chilli Fusarium wilt; fungicides do not control viruses.
- Not covered: Phytophthora blight, bacterial wilt, early blight / Alternaria leaf spot, Choanephora blight, murda complex, cucumber mosaic and other viruses, and black pepper (Piper nigrum) diseases. No sourced material of sufficient quality was retrieved for them.

Appendix B. Image-classification guide for the model classes

The table maps the pepper classes used by an image model (bacterial spot) and a healthy-leaf class to the chapters above, with visual features and the look-alikes most likely to cause confusion. Confirm field diagnoses by checking the halo, the lesion centre and leaves of different ages, because lesions can look alike in a single photo.

Class	Chapter	What it looks like	Most likely confusion	How to separate	Management pointer
Bacterial spot	1	Small, brown to yellowish-green, water-soaked, circular leaf spots with a yellow halo that coalesce; leaf yellowing and drop; raised, scabby spots on fruit (TXP, TNT, TXT).	Cercospora leaf spot; early stages of other leaf spots	Cercospora lesions have light grey centres and dark brown margins and may reach 1 cm (TP5); bacterial spots are smaller and water-soaked.	Chapter 1: clean seed first; kasugamycin 3% SL and streptocycline (V1); copper alone is weak
Healthy leaf	-	No spots, holes, powdery growth, curling or blistering. The sources in this document do not describe healthy pepper foliage.	Early disease, nutrient or pest damage	General caution, not from a cited source: curling and thickening without spots points to leaf curl virus (chapter 7); thrips and mite damage can look similar (PX2).	Nothing to treat; keep scouting nursery and field